

ChE 427 (Entrepreneurship in Chemical Engineering)
Department of Chemical Engineering
University of Illinois Chicago
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Anderson Quiz:

- 1- Motivation - The most important thing for an entrepreneur is making money.
Is to be their own Boss or independent, pursuing their passion
- 2- Mission - Potential employees would rather work for a company that has a noble mission than one that pays the highest salary.
Not always. Some people will want to work because of the salary, while others will work because the company's mission aligns with their values and purpose
- 3- Intellectual property - I have written a computer code. Should I protect my intellectual property with a trademark, patent, or copyright?
Copyright. Computer code is typically protected by copyright, not a trademark or patent.
- 4- Team Building - It is better to find co-founders who think alike, rather than think differently.
It's better to have co-founders who think differently to provide diverse perspectives and robust problem-solving.
- 5- Strategic planning - Deciding whether to have a sale next week is an example of strategic thinking.
It is not necessarily an example of strategic thinking because it can be a part of a larger strategic plan, depending on the thought process behind the decision. Perplexity
- 6- Sales/Marketing
 - If a product is good enough and marketed well enough, salespeople aren't necessary.
Not necessarily true depending on the market
 - The most important task of the marketing department is writing good ads and maintaining the website.
Not exactly. Those are tactics, but the real core of marketing is understanding the customer and market. ChatGPT
 - Product/Market Fit - An amazing product idea with limited sales is better than a boring product with massive sales.
The reverse is often true. Investors and markets value traction (sales, users, adoption) over just "amazing ideas." A "boring" product that solves a real pain point and sells consistently (e.g., payroll software, plumbing supplies) often beats a flashy product with

no buyers. Product/Market Fit means customers want what you're building so badly that sales almost "pull themselves. Without that, even the best idea fails. ChatGPT

- 7- Funding – Good investors squeeze as much money from founders to increase their return on investment.

False. Good investors seek partnership and sustainable growth, not just maximizing their immediate return by squeezing founders. Good investors want founders motivated and supported, as the startup's success depends heavily on the founders.

- 8- Accounting – What is the profit of a company with \$1,000,000 in annual revenue, fixed costs of \$100,000 per year, and variable costs of \$400,000 per year? What is the percent profit margin?

Percent profit merging = $(\text{profit}/\text{Revenue}) * 100$

Total Cost = \$100,000 + \$400,000=\$500,000

Revenue = \$1,000,000

Profit=\$1,000,000-\$500,000=\$500,000

Percent profit merging = $(\$500,000/\$1,000,000) * 100=50\%$

- 9- Finance – My company can earn \$200,000 profit per year by investing \$3,000,000. Should I borrow money for that project with an interest rate of 8%?

Invest \$3,000,000 → earn \$200,000 per year (≈6.7% return). Only borrow if expected return > cost of capital. ChatGPT

- 10- Cash flow – our company buys parts from overseas with upfront payment and 9-10 weeks lead time. It takes our production team 4 weeks to build and ship products to customers. Our COGs are 50% and labor cost is 15% of selling price, what payment terms should we request from our customers to stay cash flow positive.

Require 50% deposit at order and 50% on shipment, or full prepayment. Net terms (e.g., Net 30) extend your cash flow deficit. ChatGPT

- 11- Manufacturing – Manufacturing work force is abundant, and most manufacturing workers do not need math or computer skills.

False. Math and computer skills are increasingly important in modern manufacturing, and a skilled workforce can be scarce. Modern manufacturing (automation, robotics, quality control, lean processes) increasingly requires technical, math, and computer literacy. Abundant workforce ≠ trained workforce. Upskilling is key. Perplexity

- 12- Human Resources – Objective is to hire people at the lowest possible salary

False. The objective is to hire the right people at fair wages, not just the lowest salary, to retain and motivate staff.

13- Economics – A good way to increase profit is to raise the price

False. Raising the price is only one way (and not always the best); profit can increase by lowering costs, improving efficiency, or growing sales. A good way to increase profit is to optimize price relative to value → sometimes raising, sometimes lowering, sometimes bundling. Perplexity

14- Legal – Pick one – You need a lawyer to help you find ways around laws that restrict your business. Your lawyer needs to know the law. (You can provide the business background). A good lawyer keeps you out of trouble, not waiting until you have messed up.

A good lawyer keeps you out of trouble, not waiting until you have messed up. Lawyers should help you comply, not just find loopholes. Perplexity

15- Global business – The primary reason globalization is important is that it provides us with a source of inexpensive labor.

It is not only about cheap labor, but it's about access to new markets, resources, ideas, technology, and supply chains, as well as labor advantages. perplexity

16- Project management – To make sure a project gets finished on time you should plan for extra workers and overtime during the last month.

good project management = detailed upfront planning, milestones, and resource allocation throughout. Prevention > last-minute scrambling. Perplexity

17- Environment – If you don't pollute either air or water, you are good to go.

False. Not polluting air or water is good, but compliance includes other aspects like waste, land contamination, regulations, etc. Perplexity

18- Customer service –

- Providing good customer service is very costly.

Not necessarily. It often reduces costs (fewer returns, fewer complaints, higher efficiency). ChatGPT

- Providing good customer service improves profit

True. Satisfied customers generate repeat business and a positive reputation. Perplexity

19- Research and development – Ideas for new products usually come from PhD scientists.

False. New product ideas come from many places—customers, frontline workers, engineers, not only PhDs. Perplexity

20- Ethics – True or False – Safe design of bridges is the responsibility of the manager of engineering.

False. Safe design is a responsibility shared by all engineers, not just a manager; ethical safety is everyone's concern. Perplexity